



Islamic University of Najaf
College of Technical Engineering
Department of Computer Technical Engineering
2026-2027

Information Theory and Coding
Lecture 2

Probability Types

Asst. Lecturer: Jenan Abdul-Ridha Abbas
M.Sc. Of Communications Engineering

Course Communication

Scan the QR code or use the link to join the class:

<https://classroom.google.com/c/ODg1OTAzNjgzODU1?cjc=bysyzqop>



Lecture Description

Lecture Title	Probability Types
Lecture location	Islamic University of Najaf / College of Technical Engineering / Department of Computer Technical Engineering
Target audience	4th Stage Students\ Department of Computer Technical Engineering
Year	2026-2027
Educational methods and techniques	Computer – Electronic screen – Classroom - Board
Assessment Methods	Activities – Tasks
Time	2 hours

Lecture Objectives

After this lecture, the student will be able to:

- ✓ Define joint, and conditional probabilities.
- ✓ Calculate joint and marginal probabilities from a probability distribution.
- ✓ Apply the conditional probability formula to dependent events.
- ✓ Explain the basic concept of Bayes' Theorem.



Probability Types

1. **Joint Probability**: is the probability of event Y occurring at the same time event X occurs. Its notation is $P(X \cap Y)$ or $P(X,Y)$, which reads: the joint probability of X and Y.

$$P(X \cap Y) \text{ or } P(X,Y)$$

For independent events:

$$P(X,Y) = P(X) \times P(Y)$$

Probability Types

Example 1:

For discrete random variable, if the probability of rolling a four on one die is $P(X)$ and if the probability of rolling a four on second die is $P(Y)$.

Find $P(X,Y)$.

Solution : $P(X) = P(Y) = \frac{1}{6}$

$$P(X,Y) = P(X) \times P(Y)$$

$$= \frac{1}{6} \times \frac{1}{6}$$

$$= \frac{1}{36} = 2.8\%$$

Probability Types

Example 2:

Given the following joint probability distribution:

$$P(x, y) = \begin{array}{cc} & \begin{matrix} 0.1 & 0.25 \end{matrix} \\ \begin{matrix} 0 \\ 0.25 \end{matrix} & \begin{matrix} 0.2 \\ 0.2 \end{matrix} \end{array}$$

Solution :

$$P(x_1) = 0.1 + 0.25 = 0.35$$

$$P(x_2) = 0 + 0.2 = 0.2$$

$$P(x_3) = 0.25 + 0.2 = 0.45$$

$$P(x) = [0.35 \quad 0.2 \quad 0.45]$$

$$P(y_1) = 0.1 + 0 + 0.25 = 0.35$$

$$P(y_2) = 0.25 + 0.2 + 0.2 = 0.65$$

$$P(y) = [0.35 \quad 0.65]$$

Probability Types

2. Conditional Probability: It is happened when there are dependent events. We have to use the symbol "|" to mean "given" when there are dependent events.- $P(B|A)$ means "Event B given Event A has occurred".- $P(B|A)$ is also called the "Conditional Probability" of B given A has occurred.- And we write it as:

$$P(A|B) = \frac{\text{number of elements of A and B}}{\text{number of elements of B}} \quad \text{or} \quad P(A|B) = \frac{P(A,B)}{P(B)}$$

Probability Types

Example :

Three fair coins are tossed.

Let: A = the event of getting more heads than tails

B = the event that the first toss is a head

Find the conditional probability $P(A | B)$.

Solution :

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

$$P(A) = \{HHH, HHT, HTH, THH\} = 4/8$$

$$P(B) = \{HHH, HHT, HTH, HTT\} = 4/8$$

$$P(A \cap B) = 3/8$$

$$P(A | B) = (3/8) / (4/8) = 3/4$$

Activity

A fair coin is tossed twice.

Let:

A=the first toss is a Head

B=at least one Head occurs

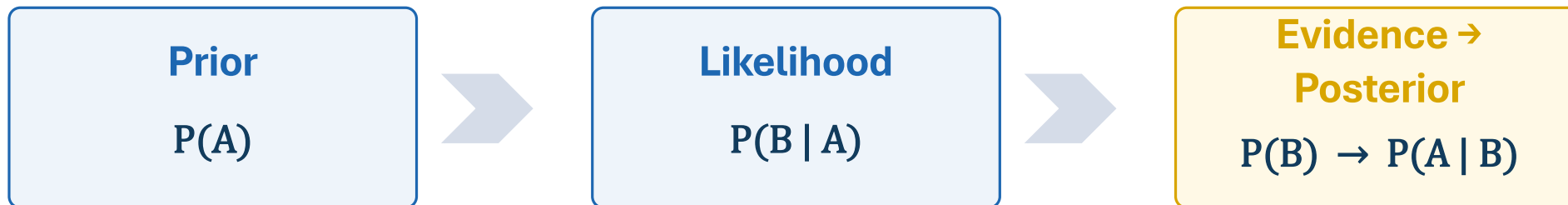
Find:

1. The joint probability $P(A \cap B)$.
2. The conditional probability $P(A | B)$.

Bayes' Theorem

Bayes' Theorem is a mathematical rule used to calculate the probability of an event based on prior knowledge of related events.

$$P(A | B) = P(B | A) P(A) / P(B)$$



Example

$$P(A) = 0.2, P(B|A)=0.05, P(B)=0.024 \rightarrow P(A|B)=0.05 \times 0.2 / 0.024 = 0.4167$$

Independent and Dependent Events

Independence means one event does not change the probability of the other.

Independent Events

A and B are independent if any equivalent condition holds:

- $P(A \cap B) = P(A)P(B)$
- $P(A|B) = P(A)$
- $P(B|A) = P(B)$

Example: successive fair coin tosses.

Dependent Events

The occurrence of one event affects the probability of another.

Example:

Selecting objects without replacement changes the probabilities on the next selection.

Test independence by comparing $P(A \cap B)$ with $P(A)P(B)$.

Homework

Q1: In a class of 100 students:

- 60 students use a laptop for studying.
- 40 students use a tablet for studying.
- 25 students use both a laptop and a tablet.

Let: A=Student uses a laptop

B=Student uses a tablet

Answer the following:

1. Calculate $P(A)$ and $P(B)$
2. Find the joint probability $P(A \cap B)$.
3. Find the conditional probability $P(A | B)$.
4. Find the conditional probability $P(B | A)$
5. Are $P(A | B)$ and $P(B | A)$ equal? Explain your answer.